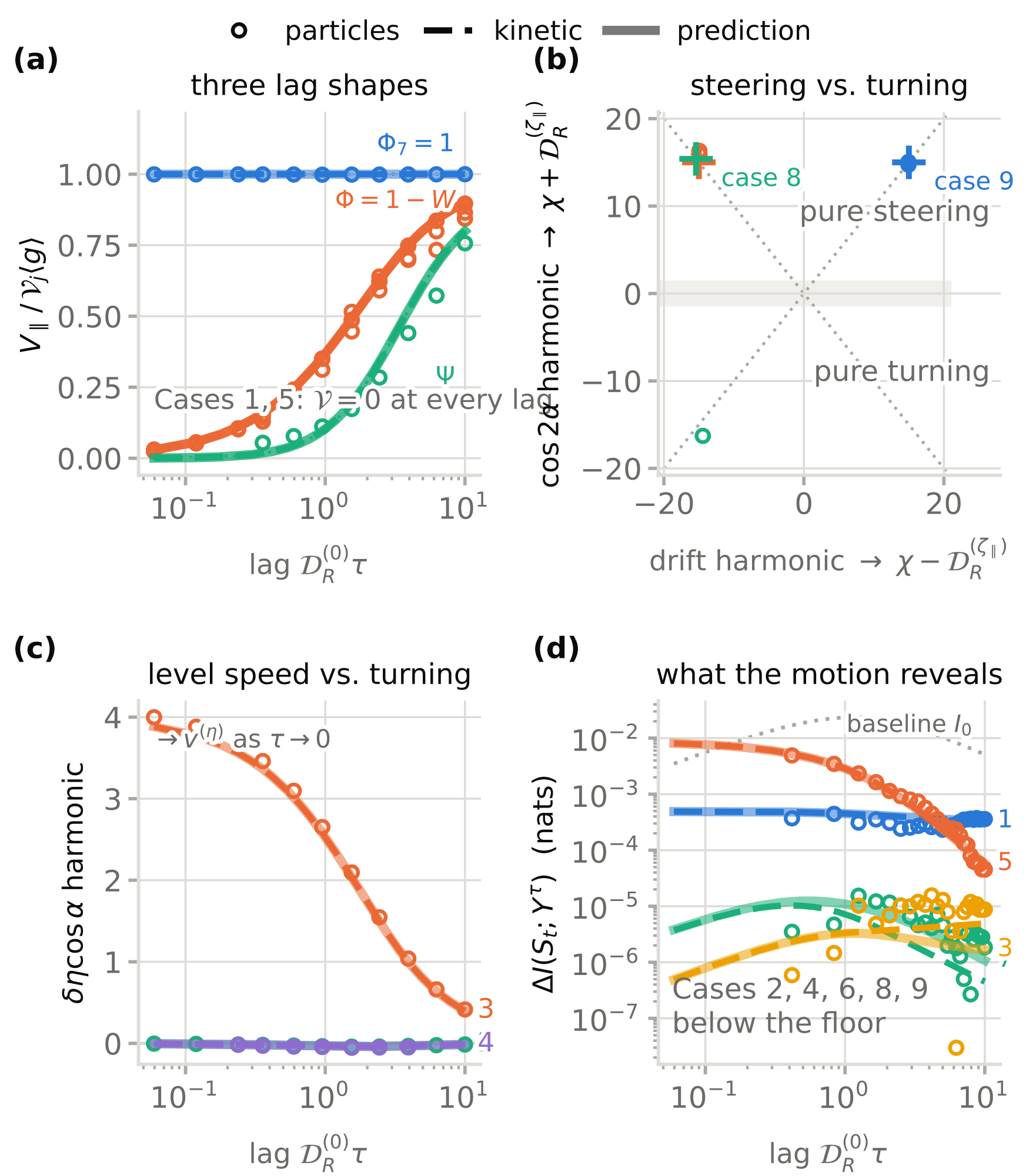

particles
kinetic
prediction
(a)
three lag shapes
$\Phi_7 = 1$
$\Phi = 1 - W$
Ψ
Cases 1, 5: $\mathcal{V} = 0$ at every lag
$V_\parallel / \mathcal{V}_j\langle g \rangle$
1.00
0.75
0.50
0.25
0.00
10^{-1}
10^{0}
10^{1}
lag $\mathcal{D}_R^{(0)}\tau$
(b)
steering vs. turning
case 8
case 9
pure steering
pure turning
$\cos 2\alpha$ harmonic $\rightarrow \chi + \mathcal{D}_R^{(\zeta_\parallel)}$
20
10
0
−10
−20
drift harmonic $\rightarrow \chi - \mathcal{D}_R^{(\zeta_\parallel)}$
(c)
level speed vs. turning
$\rightarrow v^{(\eta)}$ as $\tau \rightarrow 0$
$\delta\eta\cos\alpha$ harmonic
4
3
2
1
0
3
4
lag $\mathcal{D}_R^{(0)}\tau$
(d)
what the motion reveals
baseline I_0
$\Delta I(S_t; Y^\tau)$ (nats)
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
1
5
3
Cases 2, 4, 6, 8, 9
below the floor
lag $\mathcal{D}_R^{(0)}\tau$